

# Lecture 2

Regularization & High-Dimensional Regression

ECON6083: Machine Learning in Economics

# Today's Roadmap

| # | TOPIC                 | FOCUS                                |
|---|-----------------------|--------------------------------------|
| 1 | Failure of OLS        | Invertibility, variance explosion    |
| 2 | Ridge Regression      | L2 penalty, shrinkage, dense signals |
| 3 | Lasso & Elastic Net   | L1 penalty, sparsity, selection      |
| 4 | Economic Applications | Stock&Watson vs BCH comparison       |

**Goal:** Master regularization for high-dimensional problems

# Part 1

## The Failure of OLS in High Dimensions

# The Big Data Problem

## Traditional econometrics: "Tall" data

- $N \gg p$  (many observations, few variables)
- Example:  $N = 10,000$  individuals,  $p = 10$  controls

## Modern reality: "Wide" data

- $p \approx N$  or  $p > N$  (many variables)
- Macroeconomics: Hundreds of GDP indicators (Fred-MD)
- Microeconomics: Thousands of confounders
- Text data: Each word a feature

# OLS Recap: What Could Go Wrong?

The classic formula:

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'Y$$

Requirements:

- $X'X$  must be full rank
- Requires  $\text{rank}(X) = p$

The problem when  $p > N$ :

- Rank of  $X$  at most  $N \rightarrow X'X$  is singular
- $(X'X)^{-1}$  does not exist!
- Infinite solutions, perfect in-sample fit ( $R^2 = 1$ ), terrible out-of-sample

# Variance Explosion

**Even if  $p < N$  but  $p$  large:** Multicollinearity makes variance explode

**Via SVD:**  $X = U\Sigma V'$

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 \sum_{j=1}^p \frac{v_j v_j'}{d_j^2}$$

**If features correlated:**  $d_j \approx 0 \rightarrow \frac{1}{d_j^2} \rightarrow \infty$

**Example:** Housing starts (+50) vs Construction permits (-49) — they cancel out, but individually unstable

# The Bias-Variance Insight

**MSE decomposition:**

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Var} + \sigma_\epsilon^2$$

**OLS (Gauss-Markov):**

- Bias = 0 (unbiased)
- Variance = Minimal among unbiased estimators

**Regularization insight:** Accept small bias to drastically reduce variance!

## When Does OLS Fail? Summary

| SCENARIO            | PROBLEM                 | SOLUTION               |
|---------------------|-------------------------|------------------------|
| $p > N$             | $X'X$ singular          | Ridge: add $\lambda I$ |
| $p \approx N$       | Near-singular, unstable | Regularization         |
| Correlated features | Variance explosion      | Shrinkage              |
| Many small effects  | High variance           | Ridge                  |
| Few true effects    | Hard to identify        | Lasso                  |



# Part 2

Ridge Regression (L2 Regularization)

# The Ridge Estimator

Add L2 penalty to loss:

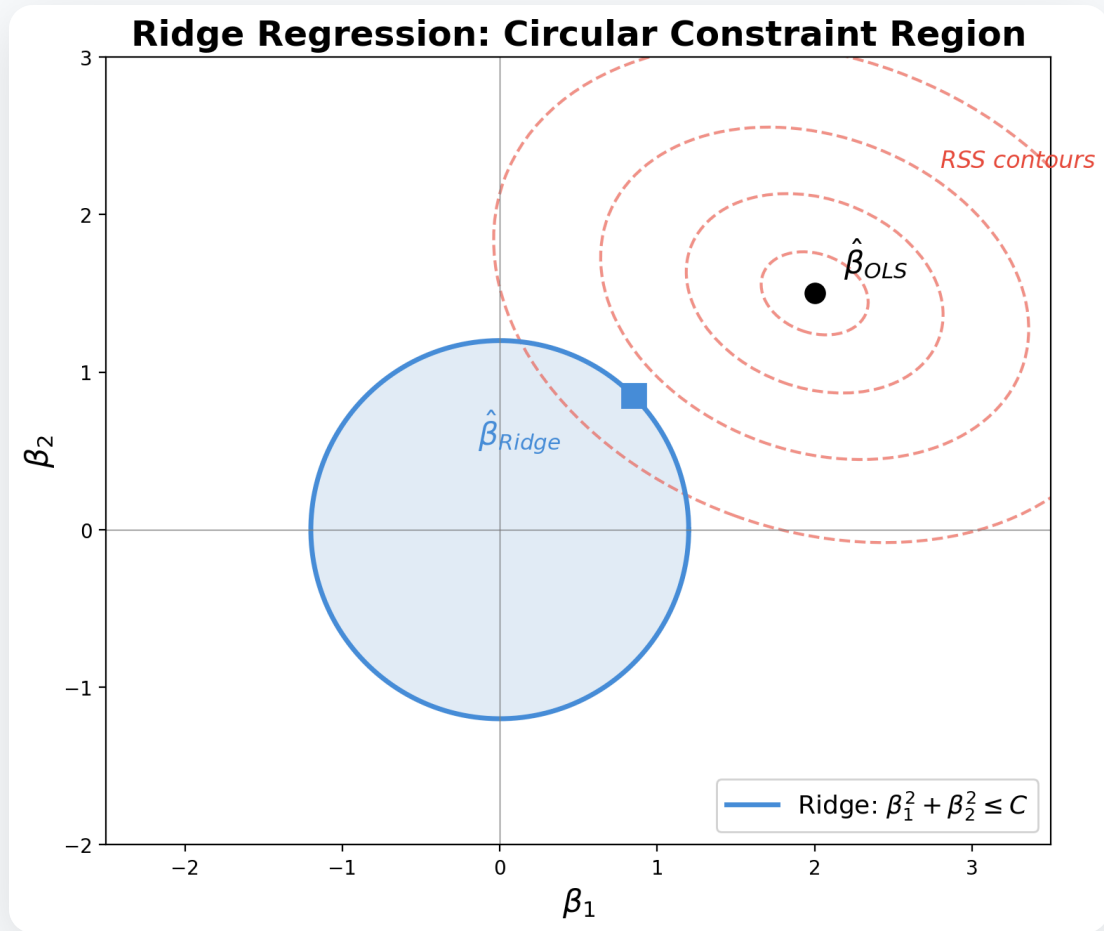
$$\min_{\beta} ||Y - X\beta||_2^2 + \lambda ||\beta||_2^2$$

Closed-form solution:

$$\hat{\beta}_{Ridge} = (X'X + \lambda I)^{-1} X'Y$$

**Key difference:**  $+\lambda I$  makes matrix always invertible, even when  $p > N$

# Geometric Interpretation



# Why Ridge Works

## Effect on eigenvalues:

- Original:  $d_j^2$  (can be  $\approx 0$ )
- Ridge:  $d_j^2 + \lambda$  (always  $> 0$ )

## Shrinkage intuition:

$$\hat{\beta}_j^{Ridge} \approx \hat{\beta}_j^{OLS} \times \frac{\text{Var}(X_j)}{\text{Var}(X_j) + \lambda}$$

| $\text{Var}(X_j)$ | SHRINKAGE   | EFFECT           |
|-------------------|-------------|------------------|
| Large             | $\approx 1$ | Keep coefficient |
| Small             | $\approx 0$ | Shrink to zero   |

# Ridge: Pros and Cons

## Advantages:

- Handles multicollinearity
- Always has solution ( $p > N$  OK)
- Stable predictions
- **Excellent for dense signals** (many small effects)

## Disadvantages:

- **No variable selection** (all  $\beta_j \neq 0$ )
- Less interpretable

# Bayesian Interpretation

Ridge equivalent to MAP estimation with Gaussian prior:

$$\beta_j \sim N(0, \sigma^2 / \lambda)$$

## Interpretation:

- Prior belief: coefficients near zero
- Strength controlled by  $\lambda$
- Data updates this belief

**Connection:** Larger  $\lambda$  = stronger prior = more shrinkage

# Choosing $\lambda$ via Cross-Validation

## K-Fold CV procedure:

1. Split data into K folds
2. For each candidate  $\lambda$ : train on K-1 folds, predict on held-out fold, compute MSE
3. Select  $\lambda$  with lowest average MSE
4. Refit on full training set

In Python: `RidgeCV` and `LassoCV` do this automatically

## Ridge in Python

```
from sklearn.linear_model import Ridge, RidgeCV
from sklearn.preprocessing import StandardScaler

# Always standardize first!
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_train)

# Cross-validation to choose lambda
ridge_cv = RidgeCV(alphas=[0.01, 0.1, 1.0, 10.0, 100.0], cv=5)
ridge_cv.fit(X_scaled, y_train)
print(f"Best alpha: {ridge_cv.alpha_}")
```

**Important:**  $\lambda$  penalizes large coefficients — different scales → unfair penalty



# Part 3

## Lasso & Elastic Net

# The Lasso Estimator

**Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996):**

$$\min_{\beta} \frac{1}{2N} \|Y - X\beta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j|$$

**Key difference from Ridge:** L1 penalty (absolute value)

**No closed-form solution** — use coordinate descent

# Soft-Thresholding: Why Lasso Yields Sparsity

**Coordinate descent solution** (standardized features):

$$\hat{\beta}_j = S(z_j, \lambda) = \text{sign}(z_j) \max(|z_j| - \lambda, 0)$$

**Three cases:**

- $z_j > \lambda \rightarrow \hat{\beta}_j = z_j - \lambda$  (shrink)
- $|z_j| \leq \lambda \rightarrow \hat{\beta}_j = 0$  (**exactly zero!**)
- $z_j < -\lambda \rightarrow \hat{\beta}_j = z_j + \lambda$  (shrink)

**Lasso performs variable selection!**

# Soft-Thresholding vs Ridge Shrinkage

## Soft-thresholding (Lasso):

- Shrink *and* select:  $S(z, \lambda) = \text{sign}(z) \max(|z| - \lambda, 0)$

## Ridge shrinkage:

- Shrink only:  $R(z, \lambda) = \frac{z}{1+\lambda}$
- Never zero — no selection

## Hard-thresholding (Best Subset):

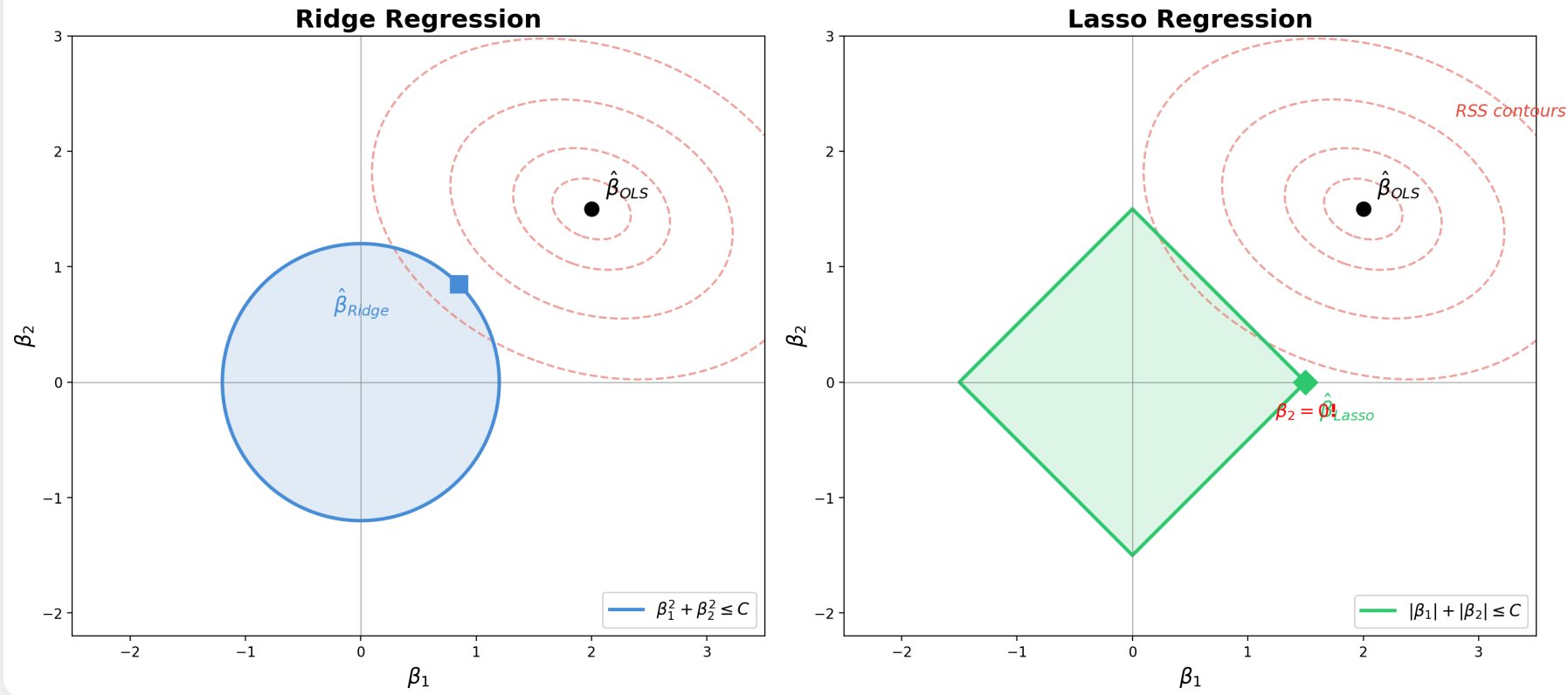
- Select only:  $H(z, \lambda) = z \cdot \mathbb{1}(|z| > \lambda)$
- Discontinuous — NP-hard to compute

## Ridge vs Lasso: Key Difference

|            | RIDGE             | LASSO              |
|------------|-------------------|--------------------|
| Operation  | Multiply by $< 1$ | Subtract $\lambda$ |
| Zero?      | Never exactly     | Can hit exactly    |
| Selection? | No                | Yes                |
| Best for   | Dense signals     | Sparse signals     |

# Geometric Intuition

## Why Lasso Produces Sparse Solutions



**Ridge:**  $\beta_1^2 + \beta_2^2 \leq C$  (circle) — smooth boundary, no corners

**Lasso:**  $|\beta_1| + |\beta_2| \leq C$  (diamond) — corners on axes  $\rightarrow$  solution likely at corner (one coefficient = 0)

# Lasso Limitations

## Saturation:

- If  $p > N$ , Lasso selects at most  $N$  variables
- Might miss important features

## Grouping problem:

- Correlated variables ( $\rho \approx 1$ )
- Lasso randomly picks one, drops others
- Economists dislike arbitrary selection

# Elastic Net: Best of Both Worlds

Combined penalty (Zou & Hastie, 2005):

$$\min_{\beta} \frac{1}{2N} \|Y - X\beta\|_2^2 + \lambda \left[ \alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right]$$

Coordinate descent update:

$$\hat{\beta}_j = \frac{S(z_j, \lambda\alpha)}{1 + \lambda(1 - \alpha)}$$

- **Numerator:** Lasso-type sparsity (soft-thresholding)
- **Denominator:** Ridge-type shrinkage



# Elastic Net Fixes Grouping

Lasso with correlated  $X_1, X_2$ : Arbitrarily picks one

**Elastic Net property:** As  $\rho \rightarrow 1$ , bound on  $|\hat{\beta}_1 - \hat{\beta}_2| \rightarrow 0$

→ Correlated variables get **similar** coefficients

| $\alpha$ | METHOD      | SPARSITY | GROUPING |
|----------|-------------|----------|----------|
| 0        | Ridge       | No       | Yes      |
| 0.5      | Elastic Net | Moderate | Yes      |
| 1        | Lasso       | Full     | No       |

# Python Cheat Sheet

```
from sklearn.linear_model import RidgeCV, LassoCV, ElasticNetCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

# ALWAYS use Pipeline to prevent data leakage
models = [
    ("Ridge", RidgeCV(alphas=[0.01, 0.1, 1, 10, 100])),
    ("Lasso", LassoCV(cv=5)),
    ("ElasticNet", ElasticNetCV(l1_ratio=[0.1, 0.5, 0.7, 0.9 ...
]

for name, model in models:
    pipe = Pipeline([("scaler", StandardScaler()), ("model", ...
    pipe.fit(X_train, y_train)
    print(f"{name}: {pipe.score(X_test, y_test):.4f}")
```

## Lasso in Python: Feature Selection Example

```
from sklearn.linear_model import LassoCV
import numpy as np

# Fit Lasso with cross-validation
lasso = LassoCV(cv=5).fit(X_train, y_train)

# Check how many features selected
n_selected = np.sum(lasso.coef_ != 0)
print(f"Selected {n_selected} out of {X.shape[1]} features")

# Identify selected features
selected_features = np.where(lasso.coef_ != 0)[0]
print(f"Best alpha: {lasso.alpha_:.4f}")
```

**Key:** Lasso returns sparse coefficients — interpretable model

# Part 4

## Economic Applications

# Application 1: Growth Regressions

**Classic problem (Sala-i-Martin, 1997):** "I Just Ran Two Million Regressions"

**Question:** What determines GDP growth?

**Challenge:**

- Theory suggests "Openness" matters
- How to measure? Tariffs? Quotas? Trade volume?
- 50+ possible proxies

**Lasso solution:** Let data select most predictive features

# Reducing Researcher Degrees of Freedom

## Traditional approach:

- Researcher tries many specifications
- Reports "best" one
- p-hacking risk

## Lasso approach:

- Pre-specify all candidates
- Algorithmic selection
- Reduces arbitrary choices

**Benefit:** More transparent, reproducible research

## Application 2: Causal Inference with High-Dimensional Controls

Setup (Belloni, Chernozhukov, Hansen, 2014):

$$Y_i = \alpha D_i + X_i' \beta + \epsilon_i$$

**Goal:** Estimate treatment effect  $\alpha$

**Problem:** High-dimensional confounders  $X$  ( $p \approx 234$ ,  $N \approx 1000$ )

**Naive Lasso mistake:** Simply run Lasso of  $Y$  on  $D, X$  — might drop confounder that predicts  $D$  but not  $Y \rightarrow$  omitted variable bias

# Post-Double Selection Lasso

**Belloni-Chernozhukov-Hansen procedure:**

**Step 1:** Lasso  $Y$  on  $X \rightarrow$  select set  $S_1$

**Step 2:** Lasso  $D$  on  $X \rightarrow$  select set  $S_2$

**Step 3:** OLS of  $Y$  on  $D$  and  $(S_1 \cup S_2)$

**Intuition:** Include variables that predict  $Y$  (reduce variance) AND variables that predict  $D$  (remove bias)



# Coordinate Descent for Lasso

**Problem:** Lasso objective is non-differentiable at zero → standard gradient descent fails

**Solution:** Coordinate descent — cycle through coordinates, optimize one at a time

**Update rule for coordinate  $j$ :**

$$\hat{\beta}_j \leftarrow S_\lambda \left( \frac{1}{N} \sum_{i=1}^N x_{ij}(y_i - \hat{y}_i^{(j)}) \right)$$

where  $S_\lambda(z) = \text{sign}(z)(|z| - \lambda)_+$  is the **soft-thresholding operator**

**Why it works:** Each coordinate-wise update has a closed form. Cycling until convergence reaches the global minimum (convex problem).

## Geometric Interpretation: Lasso vs Ridge

**Ridge ( $L_2$ ):** Constraint is a **circle** → solution touches circle at arbitrary point → rarely exactly zero

**Lasso ( $L_1$ ):** Constraint is a **diamond** → solution often touches at a corner → exact zeros → sparsity

Ridge (Circle)



Lasso (Diamond)



*"The diamond's corners align with axes → coefficients get set to exactly zero."*

# Group Lasso

**Problem:** Categorical variables with  $K$  levels create  $K - 1$  dummy variables — Lasso may select only some levels

**Group Lasso (Yuan & Lin 2006):**

$$\min_{\beta} \frac{1}{2N} \|Y - X\beta\|^2 + \lambda \sum_{g=1}^G \|\beta_g\|_2$$

- $\beta_g$ : coefficient vector for group  $g$
- $\|\beta_g\|_2$ :  $L_2$  norm of the group  $\rightarrow$  group-level sparsity

**Result:** Either **all** variables in a group are selected, or **none** are

**Economics use:** Select entire demographic blocks (age groups, education categories) rather than individual dummies

## Stock & Watson (2012) vs BCH (2014)

|                    | STOCK & WATSON (2012)       | BCH (2014)                   |
|--------------------|-----------------------------|------------------------------|
| <b>Goal</b>        | Prediction                  | Causal inference             |
| <b>Setting</b>     | 134 macro predictors        | 234 controls, gun laws       |
| <b>Best method</b> | <b>Ridge</b>                | <b>Post-double Lasso</b>     |
| <b>Signal type</b> | Dense (many small effects)  | Sparse (few key confounders) |
| <b>Key finding</b> | Ridge beats Lasso for macro | No significant crime effect  |
| <b>Lesson</b>      | Match method to DGP         | Naive selection → bias       |

**Bottom line:** Regularization is not one-size-fits-all

# Stock & Watson (2012): Why Ridge Wins for Macro

"Generalized Shrinkage Methods for Forecasting Using Many Predictors"

*Journal of Business & Economic Statistics*, 2012

**Research question:** Which regularization method best forecasts macro variables from 100+ correlated predictors?

**ML:** Compare Ridge, Lasso, PCR on FRED-MD (134 monthly series) — evaluate by MSFE

| METHOD               | MEDIAN RELATIVE MSFE |
|----------------------|----------------------|
| AR(1) benchmark      | 1.000                |
| Principal Components | 0.961                |
| Ridge                | 0.942                |
| Lasso                | 0.968                |

**Key finding:** Dense macro signal (many small effects) → Ridge dominates Lasso

# Stock & Watson: Practical Lessons

For economists doing forecasting:

1. **Don't use Lasso by default** — match method to signal structure
2. **Ridge excels when predictors share information** (common in macro)
3. **Cross-validation is essential** for choosing  $\lambda$
4. Simple benchmarks are hard to beat — regularization helps but modestly

**Broader insight:** The "best" method depends on the **data generating process**

# BCH (2014): Post-Double Selection in Action

"Inference on Treatment Effects after Selection among High-Dimensional Controls among High-Dimensional Controls"

*Review of Economic Studies*, 2014

**Research question:** Do "shall-issue" gun laws reduce violent crime, controlling for 234 potential confounders?

**ML:** Post-double selection Lasso — select controls for  $Y$  and  $D$  separately, then take union

| SPECIFICATION          | COEFFICIENT    | SIGNIFICANT? |
|------------------------|----------------|--------------|
| No controls            | +0.35          | Yes (wrong!) |
| Kitchen sink (all 234) | Unstable       | No           |
| Researcher-selected    | -0.15 to -0.80 | Varies       |
| Post-double selection  | -0.43          | No (p=0.07)  |

**Result:** No statistically significant effect — union step guards against omitted variable bias

# Why Double Selection Works

**Set  $S_1$  (predicts  $Y$ ):** Controls for outcome variation

**Set  $S_2$  (predicts  $D$ ):** Controls for confounding

**Union  $S_1 \cup S_2$ :**

- Captures confounders even if weak for  $Y$
- Maintains low variance
- Delivers consistent  $\hat{\alpha}$

**Modern standard for high-dimensional causal inference**



# Key Takeaways

# Summary: High-Dimensional Problem

## OLS fails when $p$ large:

- Singularity ( $p > N$ )
- Variance explosion (multicollinearity)

## Solution: Regularization

- Accept small bias  $\rightarrow$  drastically reduce variance
- Improve out-of-sample prediction

## Summary: Ridge vs Lasso vs Elastic Net

| FEATURE            | RIDGE (L2)               | LASSO (L1)         | ELASTIC NET                               |
|--------------------|--------------------------|--------------------|-------------------------------------------|
| Penalty            | $\sum \beta_j^2$         | $\sum  \beta_j $   | $\sum \lambda \beta_j^2 + \sum  \beta_j $ |
| Solution           | Closed form              | Coordinate descent | Coordinate descent                        |
| Sparsity           | No (all $\beta \neq 0$ ) | Yes                | Yes                                       |
| Use when           | Dense signals            | Sparse signals     | Unsure                                    |
| Macro forecasting  | Best                     | Suboptimal         | OK                                        |
| Variable selection | No                       | Yes                | Yes                                       |

# Practical Recommendations

| SCENARIO                                       | RECOMMENDED METHOD             |
|------------------------------------------------|--------------------------------|
| Macro forecasting (many correlated predictors) | Ridge or Elastic Net           |
| Growth regressions (select from many theories) | Lasso or Elastic Net           |
| Causal inference (high-dimensional controls)   | Post-Double Selection Lasso    |
| Unsure about sparsity                          | Elastic Net ( $\alpha = 0.5$ ) |

# Common Pitfalls

| PITFALL                          | SOLUTION                                     |
|----------------------------------|----------------------------------------------|
| Forgetting to standardize        | Always use <code>StandardScaler</code> first |
| Naive Lasso for causal inference | Use Post-Double Selection                    |
| Wrong regularization strength    | Cross-validate $\lambda$ (and $\alpha$ )     |
| Assuming sparsity when dense     | Use Ridge or Elastic Net                     |

## Further Reading

| PAPER                                                                          | AUTHORS                | KEY CONTRIBUTION      |
|--------------------------------------------------------------------------------|------------------------|-----------------------|
| Regression Shrinkage and Selection via the Lasso                               | Tibshirani (1996)      | Original Lasso paper  |
| Ridge Regression: Biased Estimation                                            | Hoerl & Kennard (1970) | Original Ridge paper  |
| Regularization and Variable Selection via Elastic Net                          | Zou & Hastie (2005)    | Elastic Net           |
| Generalized Shrinkage Methods for Forecasting                                  | Stock & Watson (2012)  | Macro forecasting     |
| Inference on Treatment Effects after Selection among High-Dimensional Controls | BCH (2014)             | Post-Double Selection |

# Data and Code

## This lecture:

- Slides source: `_slides/Lecture02-Regularization and High-Dimensional Regression-Slides.md`
- Exercises: `exercises/Lecture02-Exercise.ipynb`

## Key packages:

- Python: `glmnet-python`, `scikit-learn` (Lasso, Ridge, Elastic Net)

## Datasets:

- HIV drug resistance (Rhee et al. 2006); spam classification

# Thank You!

**Next Lecture:** Trees, Random Forests, and Boosting

**Reading:** Breiman (2001), "Random Forests"